Galaxy Morphology Evolution via Redshift-Conditioned Diffusion Models: Generation and Physical Evaluation

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ABSTRACT

Understanding how galaxy morphology evolves with redshift is an important problem in astrophysics. We reproduce and extend a redshift-conditioned Denoising Diffusion Probabilistic Model (DDPM) that generates galaxy images at target redshifts, and present an evaluation framework including a Convolutional Neural Network (CNN) redshift predictor and MorphCNN that back-predicts redshift and morphological parameters from both real and generated images. Our reproduction matches the original morphological metric distributions to within 2%. A CNN redshift predictor achieves $\sigma_{\mathrm{NMAD}} = 0.050$ on generated images, confirming that the DDPM has learnt physically meaningful redshift-dependent features. In addition, we introduce several extensions. We fit a proper Sérsic index to replace the original ellipticity proxy. We also propose MorphCNN, a residual CNN for predicting four morphological parameters (ellipticity, semi-major axis, Sérsic index, and isophotal area) directly from galaxy images. We also introduce new training strategies including early stopping and a learning rate scheduler. Besides, we provide several analytical improvements, such as per-bin residual analysis of CNN redshift predictions, colour evolution analysis of generated samples, strategy of sample visualisation, and a systematic comparison of different morphological measurement approaches.

Key words: galaxies: evolution – galaxies: conditional diffusion models – convolutional neural networks – photometric redshift

1 INTRODUCTION

Understanding how galaxies form and evolve across cosmic time is an important task of modern astrophysics. Galaxies undergo transformations in morphology, structure, and stellar composition over billions of years. They transition from the irregular, compact, and actively star-forming systems prevalent at high redshift to the spirals and ellipticals that dominate the local Universe (Abraham et al. 1996; Conselice 2014). Redshift, z, simultaneously encodes cosmological distance and look-back time, making it the primary axis along which this evolution is traced. A reliable, scalable connection between galaxy images and redshift is therefore essential for mapping how galaxy structure changes as a function of cosmic epoch.

There are two observational methods for measuring redshift. First, spectroscopic redshifts (z_{spec}), derived from the displacement of spectral emission and absorption lines, are highly accurate but require costly, time-intensive spectroscopy. As a result, spectroscopically confirmed samples typically number only in the hundreds of thousands. Second, photometric redshifts (z_{phot}), estimated from broadband images, scale readily to the millions or billions of galaxies observed by large imaging surveys such as the Dark Energy Survey (Abbott et al. 2022, DES), the Kilo-Degree Survey (Kuijken et al. 2019, KiDS), and the Hyper Suprime-Cam Survey (Aihara et al. 2018, 2019, HSC). This method is faster, but includes larger uncertainties than spectroscopic redshifts. Both methods have their limits, which brings a challenge in tasks with a context that require both large numbers of galaxies and accurate redshift labels simultaneously.

This project addresses the image–redshift gap from two complementary directions. The **generative** direction explores: given a target redshift, can we generate realistic galaxy images whose morphological properties reflect the evolutionary features expected at that cosmic time? Such a capability would enable systematic study of morphology evolution and provide a pathway for data augmentation. The **predictive** direction explores the inverse: given galaxy images, including real observed galaxies and synthetically generated galaxies, can we accurately recover its redshift and quantitative structural parameters? Together, these two directions form a bidirectional bridge between galaxy imaging and cosmic time.

For the generative model, we chose the Denoising Diffusion Probabilistic Model (DDPM; Ho et al. 2020), since DDPM has demonstrated considerable promise for astrophysical image synthesis (Lizarraga et al. 2024; Xue et al. 2023; Smith et al. 2022). We adopt the redshift-conditioned DDPM model architecture proposed by Lizarraga et al. (2024) (Fig. 1), with full architecture and training details described in Section 4.

A key design choice in Lizarraga et al. (2024) is to condition the DDPM on redshift as a continuous scalar rather than discretising it into bins. Previously, continuous conditioning has been explored in generative modelling more broadly, from early Conditional Generative Adversarial Nets (Mirza & Osindero 2014) to diffusion-specific guidance mechanisms (Ho & Salimans 2022) and models conditioned directly on continuous variables (Ding et al. 2024). In practice, however, galaxy generation models are usually conditioned on discretised redshift bins rather than the continuous variable itself, obscuring the gradual nature of galaxy evolution (Li et al. 2024a). To improve, we treat redshift as a continuous conditioning variable,

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allowing the model to learn a smooth, physically meaningful mapping across the full range $z \in [0, 4]$. We reproduce and extend this architecture, training on the GalaxiesML dataset (Li et al. 2024b) and evaluating the physical fidelity of generated galaxy populations via Fréchet Inception Distance (Heusel et al. 2017) and morphological metrics.

A key question in evaluating a generative model is whether it has learnt the underlying physical structure and meaning. For galaxy images, physical structure is captured by **morphological parameters**: quantitative descriptors of a galaxy's geometric and photometric properties. Following (Lizarraga et al. 2024), we use 4 parameters: *Ellipticity* measures how round or stretched-out a galaxy appears. *Semi-major axis* measures its apparent size. *Isophotal area* measures the area enclosed within a fixed brightness threshold, giving a size measure independent of how the radius is defined. *Ellipticity proxy*, a heuristic approximation of the Sérsic index. In addition to (Lizarraga et al. 2024), we fit *Sérsic index* (Sérsic 1963), to see how concentrated the light is towards the centre: $n \approx 1$ corresponds to a disc-like profile, while $n \approx 4$ corresponds to a centrally concentrated elliptical bulge. Together these five parameters encode galaxy shape, size, and structural type to evolve systematically with redshift (Conselice 2014; Domínguez Sánchez et al. 2018). Extracting these metrics from both real and generated images and comparing their distributions at matched redshifts therefore constitutes a physically interpretable test of whether the DDPM has internalised galaxy morphology evolution.

For the predictive component, following the pipeline in (Lizarraga et al. 2024) and revising the code from (Li et al. 2024a), we choose a Convolutional neural networks(CNN) structure to predict redshift from a real image set and our generated set. In addition, we choose Residual Convolutional Neural Network (ResCNN) to predict the four morphological parameters from the two sets. By comparing predictions across the two sets, we obtain a further probe of the physical fidelity of the generative model. Furthermore, it provides a test for whether a model can extract physically meaningful information from both real and generated galaxy images. A well-performing model can substitute for per-image SExtractor/SEP measurements (Barbary 2016; Bertin & Arnouts 1996) to extract redshift and morphological information. This substitution enables not only substantial computational savings, but also a differentiable, GPU-native alternative that avoids SExtractor's sensitivity, such as detection thresholds, point spread function (PSF) mismatches and deblending failures (Haigh et al. 2021; Melchior et al. 2018; Xue et al. 2023). We adopt CNN and ResNet-based architectures because CNNs are well-suited to extracting hierarchical spatial features from imaging data, and have been successfully applied to galaxy morphology classification and photometric redshift estimation (Dieleman et al. 2015; Domínguez Sánchez et al. 2018; Nishizawa et al. 2020). In particular, residual architectures (He et al. 2016) have been shown to extract complex multiband features effectively for galaxy physical-property regression (Zou et al. 2025), motivating our ResNet-style encoder for morphological parameter estimation.

2 RELATED WORK

2.1 Primary References

This work reproduces and extends Lizarraga et al. (2024), which introduced a redshift-conditioned DDPM for galaxy image synthesis and validated generated images using a CNN redshift predictor. Our experiments use the GalaxiesML (Li et al. 2024b), a dataset for machine-learning dataset that pairs HSC galaxy images with spectroscopic redshifts and morphological catalogue parameters.

2.2 Generative Models for Galaxy Imaging

Generating realistic galaxy images is a consistent exploration in astrophysics. Early deep-learning approaches adopted Generative Adversarial Networks (GANs), which can produce visually convincing images but suffer from training instability and mode collapse. More recently, score-based and diffusion models, unified under a common SDE framework (Song et al. 2021), have emerged as a more stable and higher-fidelity alternative. Smith et al. (2022) applied DDPM to galaxy image simulation, producing more photometrically realistic results than GAN-based approaches. Xue et al. (2023) applied DDPM to probabilistic galaxy image deconvolution, showing that diffusion priors capture realistic morphological structure. Li et al. (2024a) explored DDPM and Conditional Variational Autoencoder (CVAE) architectures, both conditioned on redshift, and concluded that DDPM performs better than CVAE. Lizarraga et al. (2024) extended DDPMs to continuous redshift conditioning, synthesising galaxy populations across $z \in [0, 4]$ and validating generation quality via morphological metrics and a CNN redshift predictor.

2.3 Predicting Physical Parameters from Galaxy Images

2.3.1 Photometric Redshift Estimation

Several studies have attempted to predict redshift information from galaxy images. Nishizawa et al. (2020) demonstrated photometric redshift estimation on HSC data using fitting and machine-learning models trained on five-band photometry. Jones et al. (2024) applied Bayesian Neural Networks to photometric redshift prediction for the Legacy Survey of Space and Time (LSST) survey, using predictive uncertainty to identify and filter unreliable estimates. Furthermore, both Li et al. (2024a) and Lizarraga et al. (2024) employed a CNN redshift predictor to validate that generated galaxy images carry physically meaningful redshift information.

2.3.2 Morphological Parameter Prediction

Extracting physical parameters from galaxy images using CNNs has also been explored. Dieleman et al. (2015) showed that rotation-invariant CNNs can accurately classify galaxy morphology from Sloan Digital Sky Survey (SDSS) imaging. Domínguez Sánchez et al. (2018) extended this to morphological parameter regression on SDSS galaxies using deep learning. Tuccillo et al. (2018) applied CNNs directly to surface brightness profile fitting, predicting structural parameters without explicit profile decomposition. Theobald et al. (2021) proposed a Bayesian CNN for galaxy ellipticity regression, providing uncertainty estimates alongside predictions. Ghosh et al. (2022) introduced GaMPEN, a Bayesian machine-learning framework that estimates bulge-to-total ratio, effective radius, and flux on HSC imaging with posterior uncertainty quantification.

2.3.3 This Work

While prior work has addressed photometric redshift estimation and individual morphological parameters, no existing study simultaneously targets the specific combination of four metrics (ellipticity, semi-major axis, Sérsic index, and isophotal area) on the GalaxiesML dataset, and none uses these predictions explicitly as a probe of generative model fidelity. Our CNN redshift predictor and MorphCNN fill this gap, providing a unified evaluation framework that links galaxy generation and physical parameters evaluation.

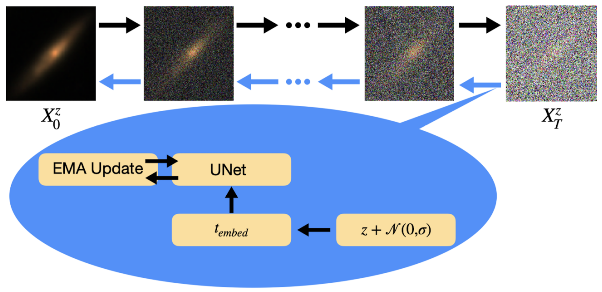

Figure 1. Overview of the redshift-conditioned DDPM (Lizarraga et al. 2024). The forward process (black arrows) gradually adds Gaussian noise to a real galaxy image X_0^z, producing a noisy image X_T^z. The reverse process (blue arrows) learns to denoise iteratively, conditioned on the redshift z perturbed by Gaussian noise $\mathcal{N}(0, \sigma)$ and embedded via $t_{\rm embed}$. The U-Net predicts the noise at each step; its weights are updated via Exponential Moving Average (EMA).

3 DATA

We use the GalaxiesML dataset (Li et al. 2024b), a machine-learning dataset built from imaging collected by the Hyper Suprime-Cam Survey Data Release 2 (Aihara et al. 2019), publicly available at `https://zenodo.org/records/11117528`. HSC is a wide-field multi-band imaging survey conducted with the 8m Subaru Telescope, covering the sky in five photometric bands (g, r, i, z, y). GalaxiesML packages the HSC galaxy images into 64×64 pixel cutouts in all five bands, paired with spectroscopic redshifts $z_{\rm spec} \in [0, 4]$ and morphological parameters derived from the HSC pipeline by running SExtractor (Bertin & Arnouts 1996) on the images. The dataset is split into a training set of 204,573 galaxies, a validation set, and a test set of 40,914 galaxies. All images and labels used in this work are drawn from GalaxiesML; the terms "HSC" and "GalaxiesML" refer to the same underlying imaging data throughout.

4 METHODOLOGY

4.1 Redshift-Conditioned Diffusion Model

4.1.1 Forward Process

A Denoising Diffusion Probabilistic Model (DDPM; Ho et al. 2020) learns to generate images by reversing a gradual noising process. The forward process adds Gaussian noise to a real galaxy image $\mathbf{x}_0$ over $T = 1000$ steps. At each step t the noisy image is:

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}\left(\mathbf{x}_t;\ \sqrt{1-\beta_t}\,\mathbf{x}_{t-1},\ \beta_t \mathbf{I}\right), \tag{1}$$

Here $\mathbf{x}_t$ is the noisy image at step t ($\mathbf{x}_0$ is the original clean galaxy image and $\mathbf{x}_T$ is pure noise), $q(\mathbf{x}_t \mid \mathbf{x}_{t-1})$ is the conditional distribution of the noisy image given the previous step, and $\beta_t \in (0, 1)$ is the noise variance added at step t, controlling how much noise is mixed in. A small β_t adds only a little noise; a larger β_t adds more. We use a linear schedule from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$, so the amount of noise grows gradually over the $T = 1000$ steps. Setting $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$, we can reach any noisy step directly:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{2}$$

which makes training efficient since any timestep can be sampled in one step.

4.1.2 Reverse Process and Training

A U-Net (Ronneberger et al. 2015), denoted $\boldsymbol{\varepsilon}_\theta$, takes the noisy image $\mathbf{x}_t$, the timestep t, and the spectroscopic redshift z as input, and predicts the noise $\boldsymbol{\varepsilon}$. The training loss is the Huber (smooth L_1) loss (Huber 1964):

$$\mathcal{L} = \mathbb{E}_{\mathbf{x}_0, \boldsymbol{\varepsilon}, t}\left[\mathcal{L}_\delta(\boldsymbol{\varepsilon},\ \boldsymbol{\varepsilon}_\theta(\mathbf{x}_t, t, z))\right], \tag{3}$$

where:

$$\mathcal{L}_\delta(y,\ \hat{y}) = \begin{cases} \frac{1}{2}(y-\hat{y})^2 & \text{if } |y-\hat{y}| \le \delta, \\ \delta\,|y-\hat{y}| - \frac{1}{2}\delta^2 & \text{otherwise,} \end{cases} \tag{4}$$

with $\delta = 1$. Huber loss is less sensitive to large noise prediction errors than MSE, which is useful for galaxy images with high dynamic range.

Once trained, new images are generated starting from pure Gaussian noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and iterating the reverse step:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\,\boldsymbol{\varepsilon}_\theta(\mathbf{x}_t, t, z)\right) + \sqrt{\beta_t}\,\boldsymbol{\varepsilon}, \tag{5}$$

where $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ for $t > 1$ and $\mathbf{0}$ at $t = 1$.

4.1.3 Redshift Conditioning and Label Smoothing

The redshift $z \in [0, 4]$ is passed to the U-Net as a continuous scalar condition. Following Lizarraga et al. (2024), a small Gaussian perturbation is added to the label during training:

$$\tilde{z} = \mathrm{clip}(z + \varepsilon_z,\ 0,\ 4), \qquad \varepsilon_z \sim \mathcal{N}(0, \sigma^2), \tag{6}$$

with $\sigma = 0.01$ in this work. The intuition behind this perturbation is that galaxies at similar redshifts should have similar appearances; by slightly randomising the conditioning label, the model is encouraged to learn a smooth mapping in which nearby redshifts share visual features, rather than treating each redshift value as a completely independent class. We investigate the effect of σ on generation quality through an ablation study in Section 5.7.

The perturbed redshift $\tilde{z}$ is incorporated into the U-Net as follows. The diffusion timestep t is first encoded into a 256-dimensional vector $\mathbf{e}_t$ via sinusoidal positional encoding (the same scheme used in transformer models). Separately, $\tilde{z}$ is projected from a scalar to the same 256-dimensional space through a learnable linear layer:

$$\mathbf{e}_z = \mathbf{W}\,\tilde{z} + \mathbf{b}, \qquad \mathbf{W} \in \mathbb{R}^{256\times 1}, \tag{7}$$

The two embeddings are then combined by element-wise addition,

$$\mathbf{e} = \mathbf{e}_t + \mathbf{e}_z, \tag{8}$$

and this combined vector $\mathbf{e}$ is injected into every encoder and decoder block of the U-Net, conditioning noise prediction on both the diffusion step and the target redshift simultaneously.

This design makes the conditioning *continuous*: $\tilde{z}$ enters as a floating-point number through a linear projection, unlike as an integer index into a discrete embedding table. As a result, the model can be conditioned on any redshift in $[0, 4]$ instead of several bins. Nearby redshift values produce similar embeddings $\mathbf{e}_z$, encouraging the network to learn a smooth, physically meaningful mapping across the redshift range.

4.1.4 Architecture and Training Details

The U-Net takes 5 input channels, one per HSC photometric band (g, r, i, z, y), and outputs 5 channels of predicted noise at 64×64 resolution.

The noise schedule uses $T = 1000$ steps with β_t increasing linearly from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$.

To stabilise training, we applied Exponential Moving Average (EMA) (Karras et al. 2024) to the U-Net model weights $\boldsymbol{\theta}$. After each gradient update, a shadow copy $\boldsymbol{\theta}_{\text{ema}}$ of the weights is maintained:

$$\boldsymbol{\theta}_{\text{ema}} \leftarrow \lambda\, \boldsymbol{\theta}_{\text{ema}} + (1 - \lambda)\, \boldsymbol{\theta}, \tag{9}$$

with decay $\lambda = 0.995$. Because $\boldsymbol{\theta}_{\text{ema}}$ changes slowly and is less affected by noisy gradient updates, it produces more stable generated images than the raw weights. The EMA model was used for all image generation.

Training used the AdamW optimiser (Loshchilov & Hutter 2019) with learning rate 2×10^{-5} and batch size 128, for up to 600 epochs on the CSD3 Ampere (NVIDIA A100 GPU) cluster.

Compared to Lizarraga et al. (2024), we introduced two training improvements. First, we added a `ReduceLROnPlateau` learning rate scheduler (reduction factor 0.5, patience 10 epochs), which halves the learning rate whenever validation loss stagnates. Second, we introduced early stopping: training was halted when validation loss did not improve for 20 consecutive epochs, or when the learning rate fell below 10^{-7}. These changes prevent overfitting and reduce unnecessary compute. We adopt $\sigma = 0.01$ for redshift label perturbation, based on our ablation study (Section 5.7).

4.2 Evaluation Methods

To verify that the DDPM has learned physically realistic galaxy morphologies, we evaluate the generated images using four approaches. First, we compare the distributions of four **morphological metrics** — ellipticity, semi-major axis, ellipticity proxy, and isophotal area — between generated images and the GalaxiesML test set, measured using SEP (Section 4.2.1). As an extension, we replace the ellipticity proxy with a proper fitted Sérsic index for a more physically meaningful comparison (Section 4.2.4). Second, we compute the **Fréchet Inception Distance (FID)** between the test set and generated images to assess overall visual similarity (Section 4.2.2). Third, we train a **CNN redshift predictor** on the GalaxiesML training set and apply it to generated images, comparing predicted and true redshifts to verify the physical accuracy of the DDPM's redshift conditioning (Section 4.2.3). Fourth, we design and train **MorphCNN** on the GalaxiesML training set and apply it to generated images to independently assess whether the correct morphological parameters are reproduced (Section 4.2.5).

4.2.1 SEP-based Morphological Metric Extraction

SEP (Source Extractor as a Python library; Barbary 2016) is a Python implementation of the SExtractor algorithm (Bertin & Arnouts 1996), which detects astronomical sources by identifying connected pixels that exceed a threshold above the local background and fits elliptical apertures to each detected source. In this work, we use SEP to extract four morphological parameters from each galaxy image:

- **Ellipticity**: a measure of how elongated a galaxy appears, calculated as

$$\epsilon = 1 - \frac{b}{a}, \tag{10}$$

where a and b are the semi-major and semi-minor axes of an ellipse fitted to the galaxy by SEP, measuring how far the galaxy's brightness extends along each axis. $\epsilon = 0$ is perfectly circular; $\epsilon \to 1$ is highly elongated.

- **Semi-major axis** a : the length of the longest axis of the galaxy's ellipse.

- **Ellipticity proxy** : a heuristic approximation of the Sérsic index, computed from the SEP-measured ellipticity ϵ using the formula (Lizarraga et al. 2024):

$$n_{\text{proxy}} = \frac{\ln 2}{\ln\left(\dfrac{1+\epsilon}{1-\epsilon}\right)}. \tag{11}$$

This is not a formal profile fit; see Section 4.2.4 for the proper fitted Sérsic index.

- **Isophotal area** : the total number of pixels belonging to the detected source (`obj['npix']`), i.e. all pixels above the 1.5σ detection threshold. It reflects the galaxy's apparent extent on the sky.

Before source detection, all images were transformed with an arcsinh stretch:

$$\mathbf{x}' = \operatorname{arcsinh}(\mathbf{x}), \tag{12}$$

which compresses bright pixel values and makes faint structures easier to detect. This pipeline was applied identically to real and generated images to ensure a fair comparison.

4.2.2 Fréchet Inception Distance (FID)

The Fréchet Inception Distance (FID; Heusel et al. 2017) measures the visual quality of generated images by comparing the distribution of Inception-v3 (Szegedy et al. 2016) features between real and generated sets:

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2\left(\Sigma_r \Sigma_g\right)^{1/2}\right), \tag{13}$$

where μ_r, Σ_r and μ_g, Σ_g are the mean and covariance of features for real and generated images respectively. A lower FID indicates that generated images are closer in appearance to real galaxies.

4.2.3 CNN Redshift Predictor

Lizarraga et al. (2024) validated generated images by testing whether a CNN trained on real galaxies could correctly recover the redshift of generated ones: if the predictor succeeds, it implies the generated images carry physically realistic redshift-dependent features. We reproduce this evaluation using the dual-branch CNN predictor from Li et al. (2024a), extending it with the Normalised Median Absolute Deviation (NMAD; equation 15) and residual plots across the full range $z \in [0, 4]$.

The predictor uses a dual-branch architecture (Li et al. 2024a): a CNN branch that extracts morphological features from the galaxy image, and an MLP branch that extracts photometric colour information from broadband magnitudes. The image branch is a six-layer convolutional network that maps the $64 \times 64 \times 5$ input to a 32-dimensional feature vector. The magnitude branch is a six-layer fully connected network that takes the five HSC band magnitudes (g, r, i, z, y) as input and outputs a 200-dimensional feature embedding. The two outputs are concatenated and passed through a linear layer to predict $z_{\rm photo}$.

The model is trained using the HSC photometric redshift loss (Nishizawa et al. 2020):

$$\mathcal{L}_{\rm HSC} = 1 - \frac{1}{1 + \left(\frac{z_{\rm photo} - z_{\rm spec}}{0.15}\right)^2}, \quad (14)$$

which is bounded in $[0, 1]$ and penalises large outliers less aggressively than MSE. The predictor is trained for 20 epochs on the 204,573-image GalaxiesML training set, using spectroscopic redshifts $z_{\rm spec}$ as ground truth. Then, the predictor is evaluated on the GalaxiesML test set (40,914 images), which establishes a baseline for predictor accuracy on real galaxies; and applied on the DDPM-generated set (10,000 images), where a successful redshift recovery indicates the generated images have learned physically meaningful features.

We adopt the normalized median absolute deviation (NMAD), following Ilbert et al. (2009):

$$\mathrm{NMAD} = 1.4826 \times \mathrm{median}\left(\left|\frac{z_{\rm photo} - z_{\rm spec}}{1 + z_{\rm spec}}\right|\right). \quad (15)$$

Unlike the standard deviation, NMAD is insensitive to a small number of severely mispredicted redshifts, providing a more reliable measure of typical model performance.

4.2.4 *Extension: Fitted Sérsic Index*

The original paper used an ellipticity-based proxy for the Sérsic index (Lizarraga et al. 2024), described as 11, which is a heuristic approximation.

The true Sérsic profile (Sérsic 1963) describes radial surface brightness as:

$$I(r) = I_e \exp\left\{-b_n \left[\left(\frac{r}{r_e}\right)^{1/n} - 1\right]\right\}, \quad (16)$$

where r_e is the half-light radius, I_e is the surface brightness at r_e, n is the Sérsic index, and b_n is a normalisation constant that ensures r_e encloses half the total luminosity. Values of $n \approx 1$ indicate disc-like profiles; $n \approx 4$ indicates elliptical galaxies.

We replace the proxy with a proper profile fit. The fitting procedure is as follows.

Step 1: Profile extraction. For each galaxy, a 1D radial surface brightness profile $\{(r_i, I(r_i))\}$ is extracted from the background-subtracted image using `photutils.isophote.Ellipse` (Bradley et al. 2023). To keep quality, only isophotes with positive intensity and a converged fit are kept. Galaxies with fewer than 5 valid isophotes are excluded.

Step 2: Linearisation. Given $\{(r_i, I(r_i))\}$ extracted in Step 1, the free parameters to be fitted are b_n, r_e, and n. To simplify, we take the natural logarithm of equation (16) and absorb b_n and r_e into two composite constants A and B:

$$\ln I(r) = A - B \cdot r^{1/n}, \quad (17)$$

where $A = \ln I_e + b_n$ and $B = b_n / r_e^{1/n}$ absorb the profile parameters, and n is the Sérsic index we want. Fitting this linearised form avoids the need to evaluate b_n explicitly.

Step 3: Optimisation. We fit A, B, and n by minimising the sum of squared residuals:

$$\chi^2 = \sum_i \left[\ln I(r_i) - \left(A - B \cdot r_i^{1/n}\right)\right]^2, \quad (18)$$

using `scipy.optimize.curve_fit` (Virtanen et al. 2020), which applies the Levenberg–Marquardt algorithm. This method adaptively interpolates between gradient descent and Newton's method, making it robust to poor initial guesses. The index n is constrained to $[0.1, 10]$. The fitting is run on the icelake partition (CPU) with 4 cores on CSD3.

4.2.5 *Extension: MorphCNN as Evaluation Tool*

The CNN redshift predictor demonstrates that it is possible to back-predict a physical quantity from generated images, providing an independent check on what the DDPM has learned. Inspired by this, we extend the idea to morphological parameters: rather than back-predicting redshift, can we back-predict ellipticity, semi-major axis, isophotal area, and Sérsic index directly from galaxy images? This serves two purposes. First, it provides an independent evaluation of whether the DDPM generates images with correct morphological properties, complementing the SEP-based distribution comparison. Second, it explores whether machine learning can serve as a fast, scalable alternative to traditional aperture-fitting tools such as SEP for extracting morphological parameters from galaxy images. We trained a CNN for this task, which we call MorphCNN, incorporating residual blocks (He et al. 2016) to improve gradient flow. The architecture is shown in Fig. 2.

The encoder has five stages; each stage applies a strided convolution (stride 2) followed by a residual block, halving the spatial size at each step. Global Average Pooling (GAP) reduces the feature map to a fixed-length vector, which is passed through a two-layer fully connected head with dropout ($p = 0.3$) to produce four outputs. The total parameter count is 3.76 million.

Each target parameter was log-transformed and standardised before training (see the Results section for the parameter distributions and justification). The model was trained with Huber loss (equation 4, $\delta = 1$) on standardised targets, with input images arcsinh-stretched (equation 12) and standardised per channel.

5 RESULTS

5.1 DDPM Model

The model converged after 233 epochs under early stopping. The learning rate was reduced to 1.95×10^{-8} by epoch 175, after which training was halted as the loss no longer improved (Fig. 3). The best validation loss of 7.5×10^{-3} was achieved at epoch 67, and we use the model weights from this epoch.

5.2 Generated Sample Visualisation

Figure 4 shows a visual comparison of real GalaxiesML galaxies and our DDPM-generated counterparts across a range of redshifts, extending Figure 6 of Lizarraga et al. (2024) with a revised sample selection strategy. For each target redshift, the candidate pool consists of the 100 images (from both the test set and the generated set) with the closest actual redshift. Each candidate is ranked by a quality score:

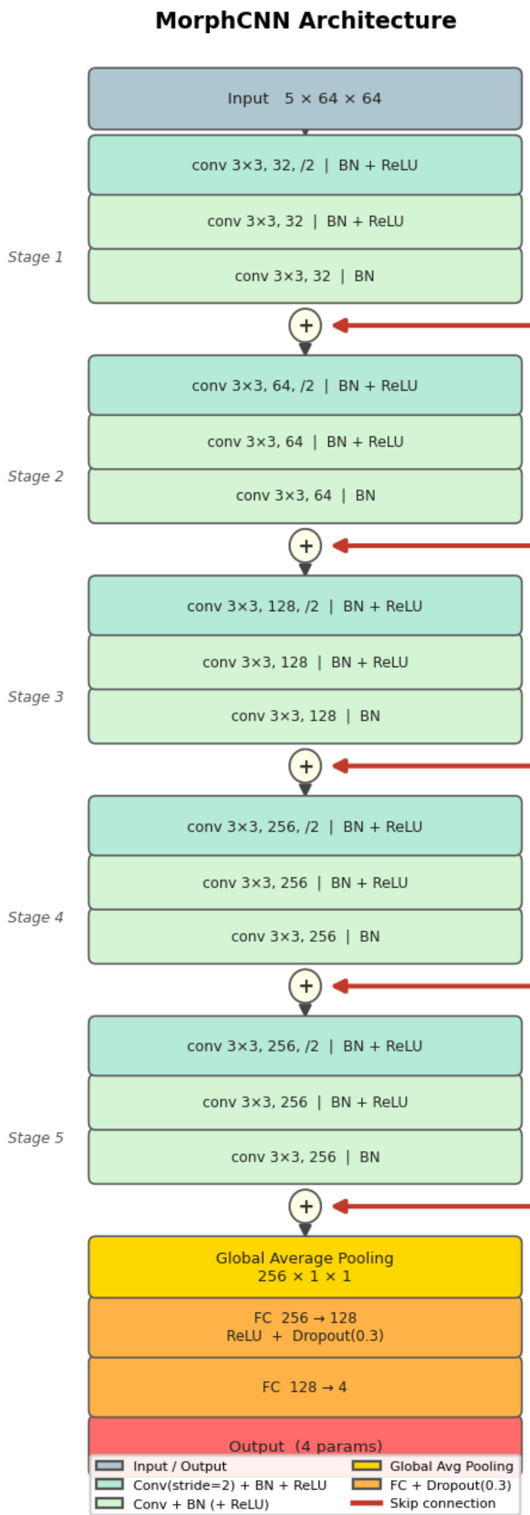

Figure 2. Architecture of MorphCNN. The input is a $5 \times 64 \times 64$ multi-band galaxy image. Five encoding stages each halve the spatial resolution via a strided convolution (stride 2) followed by a residual block (He et al. 2016). Red arrows show the skip connections within each residual block. Global Average Pooling (GAP) collapses the spatial dimensions, and a two-layer FC head with Dropout ($p = 0.3$) produces four morphological parameter predictions.

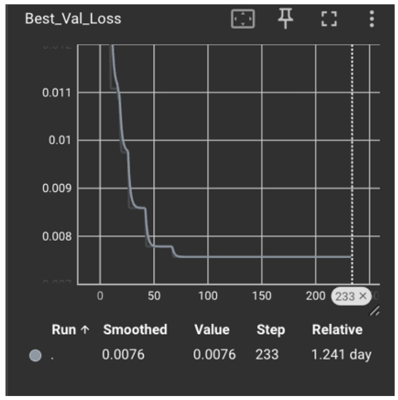

Figure 3. Validation loss curve during DDPM training. The best validation loss of 7.5×10^{-3} was achieved at epoch 67. The learning rate scheduler reduced the rate to ~0 by epoch 175, triggering early stopping at epoch 233.

$$s = \mathrm{SNR} \times center \times \sqrt{extent}, \tag{19}$$

where $\mathrm{SNR} = x_{\mathrm{peak}}/\sigma_g$ is the ratio of the brightest pixel value x_{peak} to the standard deviation σ_g of all pixel values in the g band, a measure of ratio of peak to background noise; $center = 1/(1+d/10)$ where d is the distance between the brightest pixel and the center, so $center$ decreases as the brightest pixel moves further from the image centre; and $extent$ is the number of pixels exceeding 5% of the peak pixel, quantifying how spatially extended the source is. Taking $\sqrt{extent}$ prevents large galaxies from dominating the score regardless of their brightness or centring. This criterion selects bright, well-centred, and spatially extended galaxies, allowing us to assess whether the model has learned the morphological features that evolve with redshift.

We select the 30 highest-scoring galaxy pairs (Fig. 4). At low redshift, the DDPM successfully captures the spiral morphology of disc galaxies. As redshift increases, both the GalaxiesML test-set and generated galaxies become smaller in angular size and redder in colour, because cosmological redshift shifts galaxy spectra to longer wavelengths. The DDPM reproduces both trends, indicating that the model has learned physically realistic redshift-dependent morphological and photometric evolution.

5.3 Colour Evolution

The sample visualisation in Section 5.2 shows that generated galaxies become progressively redder with increasing redshift. To investigate this colour evolution more quantitatively, we computed the $g-i$ and $r-i$ colour proxies for both the real test set and the generated images (Fig. 5).

The generated images closely reproduce the real colour–redshift trends, including the characteristic peak at $z \approx 0.5$–0.7. This trend arises from two physical effects. First, the K-correction (Hogg et al. 2002): as redshift increases, different rest-frame wavelengths are shifted into the observed bandpasses, and the 4000 Å spectral break enters the g band near $z \approx 0.5$, suppressing g-band flux and producing the observed colour peak. Second, a galaxy's spectral energy distribution, and hence its observed colours, is determined by its star formation history (Bruzual & Charlot 2003). Therefore, galaxies at different redshifts have different intrinsic colours due to their varying stellar populations and star formation histories, contributing to the scatter around the median trend.

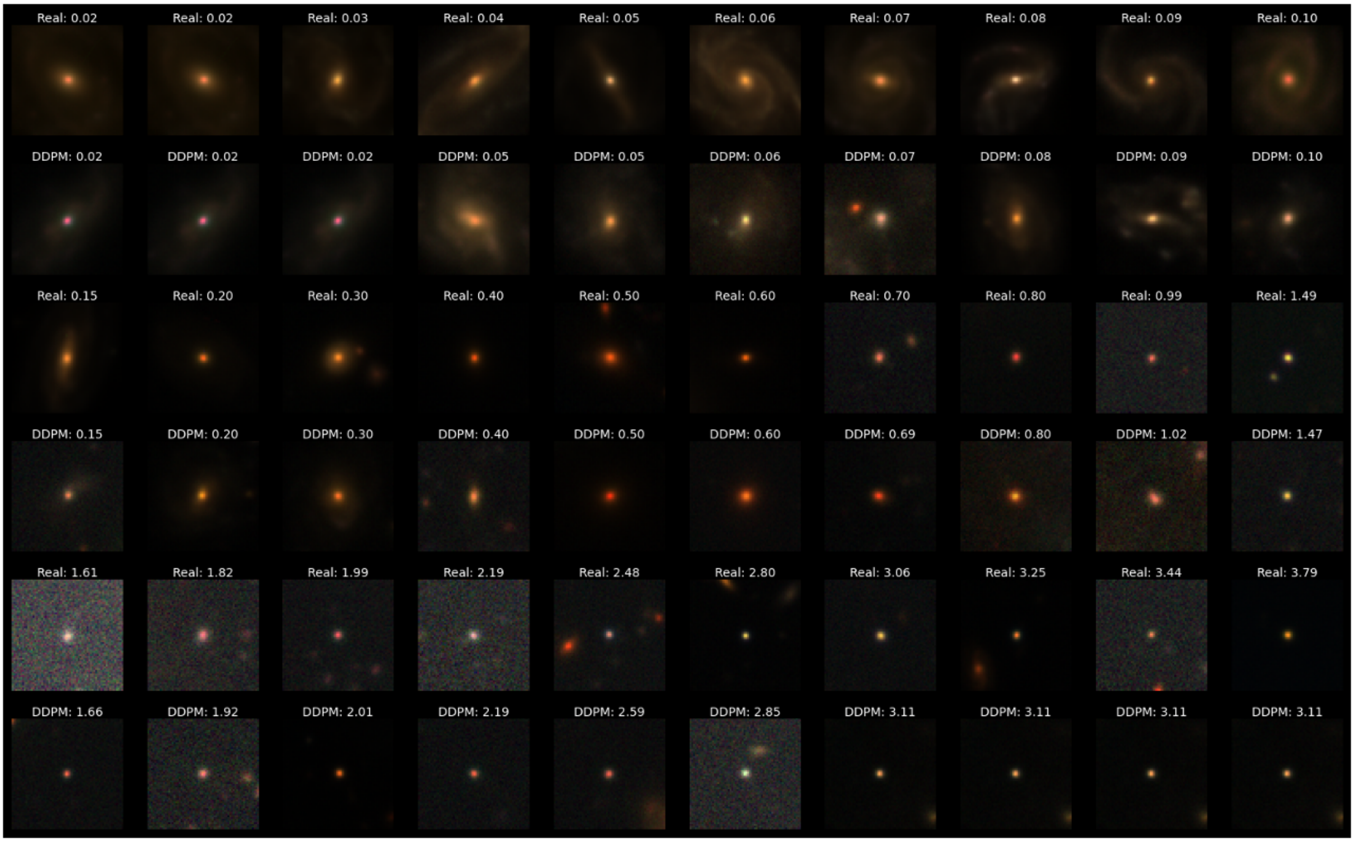

Figure 4. Visual comparison of real and synthesised galaxies across redshifts. Odd rows show GalaxiesML test-set images (real reference); even rows show corresponding DDPM-generated images at matched redshifts. Samples were selected by maximising equation (19) among images closest to each target redshift. The model successfully captures redshift-dependent visual characteristics variation.

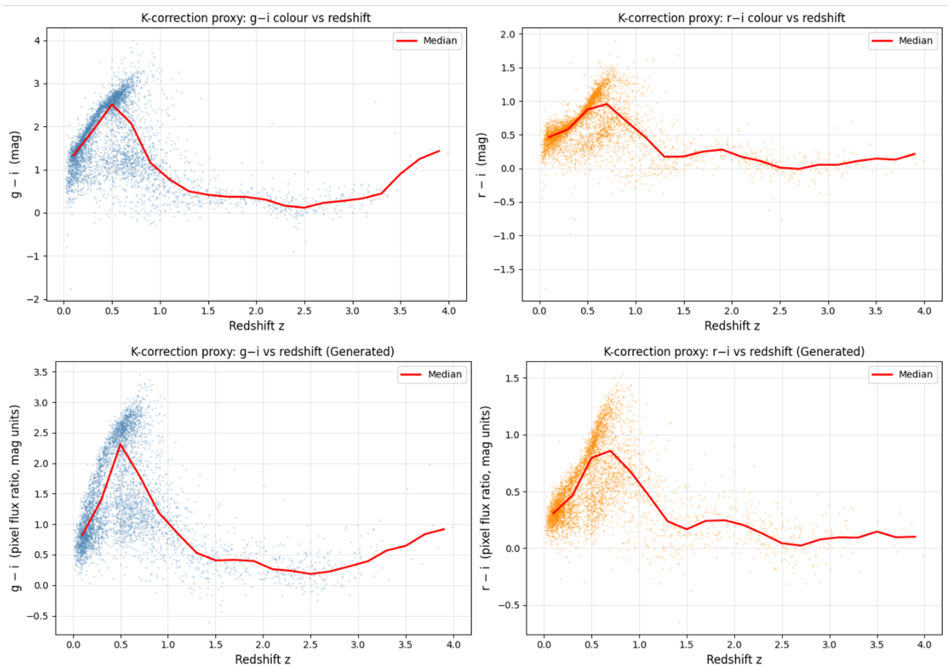

Figure 5. $g-i$ (left) and $r-i$ (right) colour proxies versus redshift for real GalaxiesML test galaxies (top) and DDPM-generated galaxies (bottom). Red lines show the median trend. The DDPM reproduces the characteristic colour peak at $z \approx 0.5$–0.7 driven by the K-correction, as well as the scatter from intrinsic galaxy colour variation.

The DDPM reproduces both the median trend and the scatter, indicating that the model has implicitly learned the colour evolution of the galaxy population as a function of redshift.

5.4 Morphological Metrics

In addition to sample visualisation, we quantitatively evaluate whether the full generated set reproduces the correct morphological distributions. Figures 6 and 7 compare the distributions of five morphological metrics — ellipticity, semi-major axis, isophotal area, ellipticity proxy, and fitted Sérsic index — between the GalaxiesML test set and our reproduced DDPM images, and additionally overlays the original paper's generated images for reference.

Table 1 shows that all Reproduce/Paper ratios within 2 per cent of unity, confirming a reliable reproduction of Lizarraga et al. (2024). Meanwhile, high paper/real ratio and reproduce/real ratio not only shows the reproduced distributions closely match the real data, also verifies the accuracy of Lizarraga et al. (2024) generation images.

Figure 8 further shows the mean of each metric as a function of redshift, confirming that the DDPM reproduces not only the overall distributions but also the correct redshift-dependent trends.

5.5 CNN Redshift Predictor

To verify that the DDPM's redshift conditioning is physically meaningful, we applied the CNN redshift predictor (Section 4.2.3) to both the GalaxiesML test set and our generated images. Figure 9 shows the photo-z scatter plot (predicted versus true/conditioned redshift) for both sets, reproducing Figure 5 of Lizarraga et al. (2024) and extending it with a bottom row of normalised residuals $\Delta z/(1+z)$ per redshift bin and NMAD (equation 15) to make systematic biases more explicit.

In both the real and DDPM panels, predictions cluster tightly around the parity line ($y = x$), confirming that the CNN successfully recovers the redshift from image features. On the GalaxiesML test set, the predictor achieves $\sigma_{\mathrm{NMAD}} = 0.017$, establishing a strong baseline. On the DDPM-generated images, $\sigma_{\mathrm{NMAD}} = 0.050$, indicating that the conditioned redshift is recoverable with reasonable accuracy, and confirming that the DDPM has learned physically meaningful redshift-dependent image features. Notably, the bottom-right panel of Fig. 9 shows that at $z \approx 0.5$–1.5, the mean prediction error on DDPM images is *lower* than on real GalaxiesML galaxies. This suggests the DDPM generates particularly "typical" galaxies images in this redshift range, and the generated images match the appearance of the galaxies that the CNN has learned in each redshift range.

However, this could also indicate reduced diversity in the generated images: if the DDPM tends to generate very similar images at each redshift (i.e. mode averaging), predictions become easier, but the generated set may under-represent the true variety of galaxy morphologies. This potential lack of diversity is discussed further in Section 6.

In addition, two other limitations are visible in the DDPM panel. First, predictions are more concentrated at low redshift compared to the real test set, suggesting the model under-generates high-redshift diversity. Second, a horizontal stripe is visible near $z \approx 2.7$, indicating a tendency to predict a fixed low value for some high-redshift inputs. These issues are discussed further in Section 6.

5.6 MorphCNN: Morphological Parameter Prediction

5.6.1 *Training*

MorphCNN is trained on the GalaxiesML training set for 22 epochs, achieving a best validation loss of 3.51×10^{-2} under Huber loss on standardised targets. The four morphological parameters — ellipticity, semi-major axis, Sérsic index, and isophotal area — are log-transformed (except ellipticity) and standardised before training, with the normalisation statistics saved to allow consistent denormalisation at inference time.

5.6.2 *Performance on the GalaxiesML Test Set*

Figure 10 shows the parity plots (predicted versus true) for all four parameters on the held-out GalaxiesML test set. Predictions cluster tightly around the identity line ($y = x$) across the full dynamic range of each parameter, with high coefficients of determination ($R^2 = 0.88$, 0.97, 0.82, 0.96 for ellipticity, semi-major axis, Sérsic index, and isophotal area respectively) and low scatter ($\sigma_{\mathrm{NMAD}} =$ 0.033, 0.226, 0.094, 27.0). The fraction of outliers exceeding 20 per cent relative error is 38.8, 5.1, 11.7, and 12.7 per cent respectively. These results confirm that MorphCNN learns a reliable mapping from multi-band galaxy images to morphological parameters.

5.6.3 *Performance on DDPM-Generated Images*

Figure 11 shows the MorphCNN-predicted distributions for both the real test set and the generated images. The two distributions match closely across all four parameters, indicating that the model assigns physically plausible morphological values to generated images.

However, when individual predictions are compared against SEP-measurements of 4 metrics on the DDPM generation set (Fig. 10, middle column), performance degrades substantially relative to the GalaxiesML test set. While a positive trend is visible, the scatter is considerably larger: σ_{NMAD} rises to 0.172, 1.965, 0.786, and 200.5 for ellipticity, semi-major axis, Sérsic index, and isophotal area respectively, and the fraction of outliers exceeding 20% climbs to 73–93%. The residual panels further reveal a systematic underestimation bias that grows with true value, particularly for Sérsic index and isophotal area.

5.6.4 *Performance gap diagnosis*

To investigate the origin of this performance gap, we first examine the CNN encoder's internal representation of the two image sets. Figure 12 shows a t-SNE projection of the 256-dimensional feature vectors extracted from the MorphCNN encoder for 1,000 randomly sampled real and generated images. The two populations mix together in feature space with no clear separation, indicating that the CNN does not perceive the two datasets as structurally different at the level of its learned representations.

Despite the absence of a feature-space domain shift, a clear photometric difference exists between the two datasets. Figure 13 shows the azimuthally averaged radial brightness profile in the g band: DDPM-generated galaxies are systematically fainter than real ones at all radii, while exhibiting a similar profile slope. Furthermore, table 2 quantifies this using signal-to-noise ratio (SNR). Across all five photometric bands, the DDPM images have SNR values significantly lower than corresponding real images, with the greatest deficit in the g and y bands.

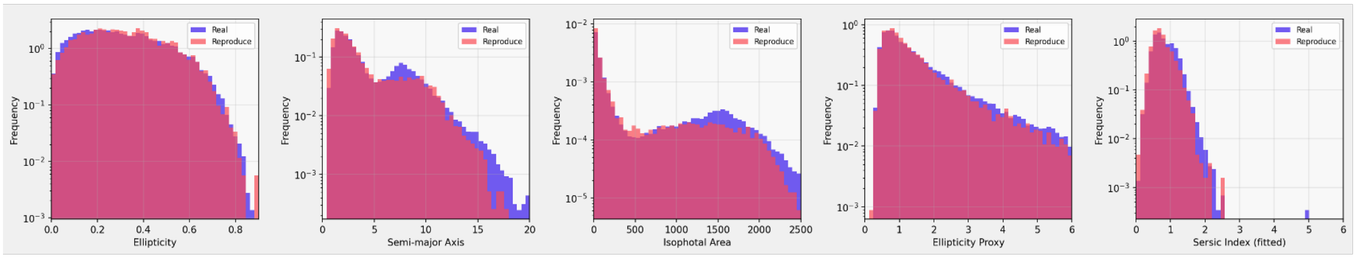

Figure 6. Morphological metric distributions (two-way comparison): real GalaxiesML test images (blue) versus our reproduced DDPM images (red). The fitted Sérsic index (rightmost panel) is an extension not present in Lizarraga et al. (2024).

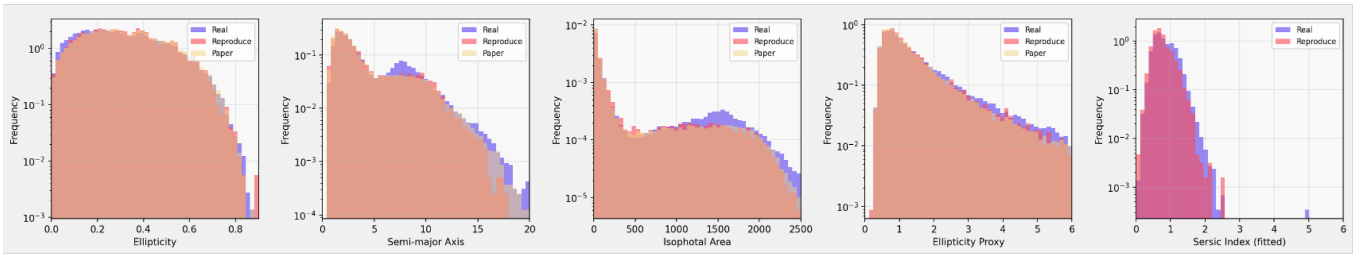

Figure 7. Morphological metric distributions (three-way comparison): real GalaxiesML test images (blue), our reproduced DDPM images (red), and the original paper's generated images (yellow).

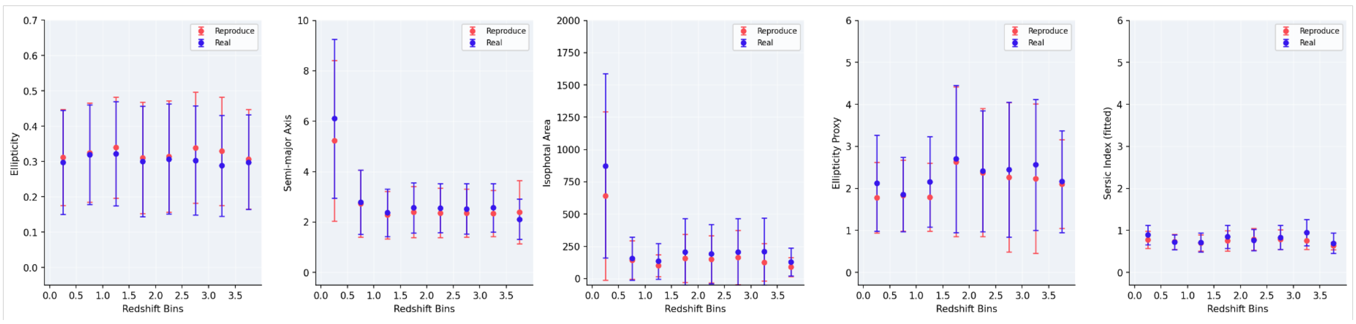

Figure 8. Mean morphological metrics as a function of redshift, comparing real GalaxiesML test galaxies (blue) and DDPM-generated galaxies (red) across redshift bins. Error bars represent 95 per cent confidence intervals on the mean. The DDPM accurately reproduces the observed redshift-dependent trends in ellipticity, semi-major axis, isophotal area, ellipticity proxy, and fitted Sérsic index.

Table 1. Morphological metrics for the GalaxiesML test set and generated images. $\bar{x}_{\rm real}$ is the real test-set mean. Paper/Real and Reproduce/Real are the ratios of the generated mean to the real mean for Lizarraga et al. (2024) and our reproduction respectively; a ratio of 1 indicates perfect agreement. Reproduce/Paper shows how closely our reproduction matches the original paper's generated images. All Reproduce/Paper ratios are within 2 per cent of unity, confirming a reliable reproduction. The fitted Sérsic index is an extension unique to this work and was not computed in Lizarraga et al. (2024) (N/A).

Metric	$\bar{x}_{\rm real}$	Paper/Real	Reproduce/Real	Reproduce/Paper
Ellipticity	0.307	1.057	1.037	0.981
Semi-major axis	4.503	0.881	0.890	1.010
Isophotal area	536.3	0.759	0.757	0.998
Ellipticity proxy	2.057	0.901	0.896	0.995
Sérsic index (fitted)	0.837	N/A	0.911	N/A

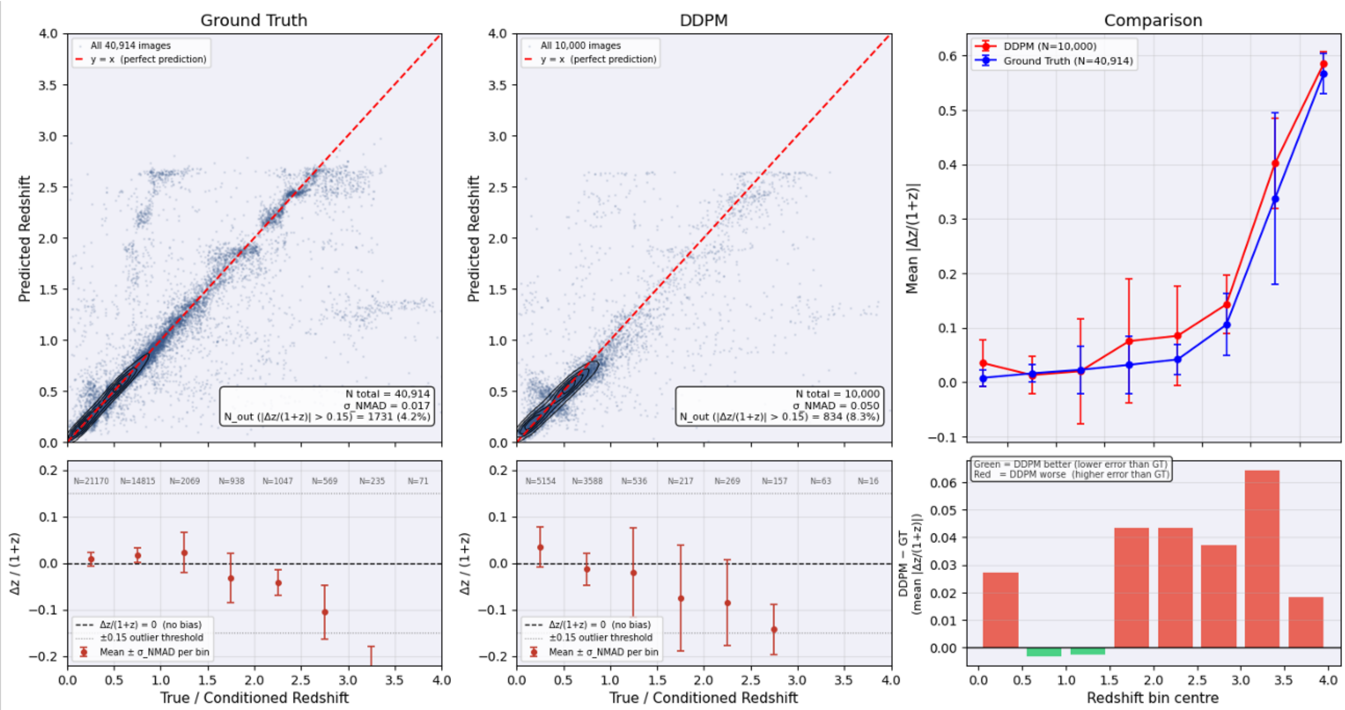

Figure 9. Photo-z scatter plots and residuals, extending Figure 5 of Lizarraga et al. (2024) with an additional bottom row. *Top left*: predicted versus true spectroscopic redshift for the GalaxiesML test set (ground truth; $N = 40{,}914$; $\sigma_{\mathrm{NMAD}} = 0.017$). *Top middle*: predicted versus conditioned redshift for DDPM-generated images ($N = 10{,}000$; $\sigma_{\mathrm{NMAD}} = 0.050$). *Top right*: mean $|\Delta z/(1+z)|$ per redshift bin for real (blue) and generated (red) images. *Bottom left and middle*: mean normalised residual $\Delta z/(1+z)$ per redshift bin for real and generated images respectively. *Bottom right*: difference in mean $|\Delta z/(1+z)|$ between DDPM and ground truth per bin (green = DDPM lower error, red = DDPM higher error).

Table 2. Signal-to-noise ratio (SNR) comparison between the GalaxiesML test set and DDPM-generated images, measured per photometric band on 500 randomly sampled images each. Signal = mean pixel value within radius $r < 10$ px; Noise = standard deviation of pixels at radius $r > 25$ px. The DDPM images have consistently lower SNR across all bands.

Band	Real SNR		DDPM SNR	
	mean	median	mean	median
g	16.90	7.36	10.32	5.26
r	21.92	12.70	14.54	8.73
i	25.51	14.82	18.99	11.36
z	20.99	13.94	14.56	9.12
y	14.97	9.89	9.58	5.62

To test whether this SNR difference causes the MorphCNN performance gap, we selectively brighten the central region of each generated image. We define the inner aperture as the circular region within $r < 10$ px of the galaxy centre. For each of 500 randomly sampled real test images, we compute the mean pixel value within this aperture, and take the median across all images as the target brightness $\bar{S}_{\mathrm{real}}$. For each generated image i with inner-aperture mean $\bar{s}_i$, we then compute a per-image scale factor:

$$k_i = \mathrm{clip}\left(\frac{\bar{S}_{\mathrm{real}}}{\bar{s}_i},\ 1,\ 5\right), \tag{20}$$

where the lower bound of 1 ensures we only ever brighten (never darken) a generated image, and the upper bound of 5 prevents extreme amplification of images where the galaxy core is near-zero — such images would otherwise be scaled by an arbitrarily large factor. The brightening is then applied via a Gaussian aperture mask ($\sigma = 10$ px) centred on the galaxy, so that the amplification is concentrated on the bright central region and the background pixels are barely affected. This raises the galaxy core brightness of each generated image to match the typical brightness of a real galaxy, while leaving the background largely unaffected. Unlike a global brightness scaling, which raises signal and noise equally and leaves SNR unchanged, our brightening approach concentrates the amplification on the bright central region, increasing SNR. Table 3 compares σ_{NMAD} before and after brightening for all four parameters. Three of the four parameters show a reduction in scatter, with the largest improvement for isophotal area (−6.1 %), suggesting that the SNR deficit does contribute to the prediction gap. However, the improvements are small and the semi-major axis error slightly increases, indicating that SNR is not the sole cause.

Table 3. σ_{NMAD} comparison between original and Gaussian-brightened DDPM images (evaluated against SEP measurements). Negative Δ indicates improvement. Three of the four parameters improve after brightening.

Parameter	Original	Brightened	Δ (%)
Ellipticity	0.1728	0.1715	−0.7
Major Axis (arcsec)	1.9651	2.0075	+2.2
Sérsic Index n	0.7863	0.7811	−0.7
Isophotal Area (px^2)	200.5	188.3	−6.1

The underlying causes of the performance gap are likely multiple and hard to explicitly explore and explain. They may include model learning reasons, differences in the pixel intensity distribution, and

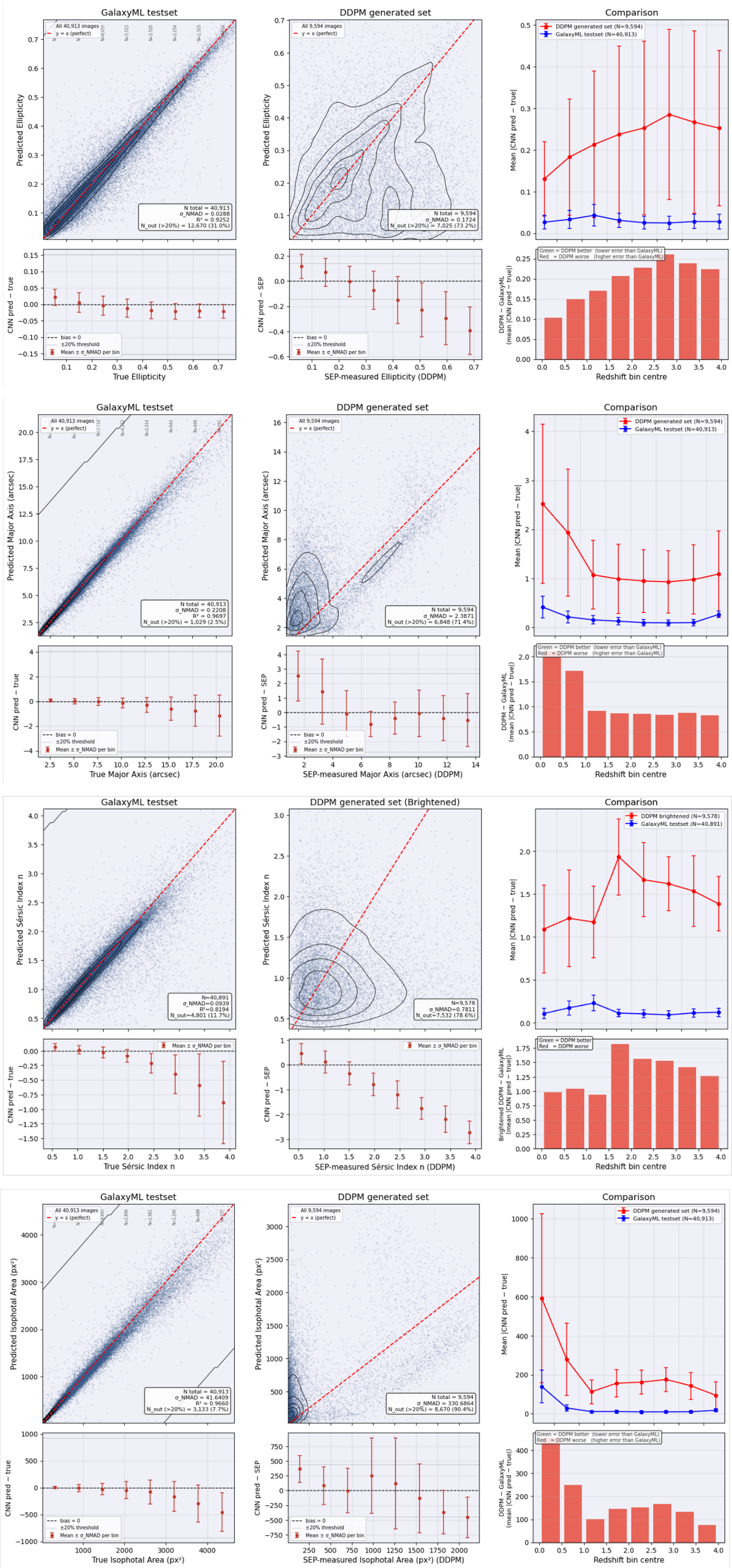

Figure 10. MorphCNN evaluation for all four morphological parameters (rows: ellipticity, semi-major axis, Sérsic index, isophotal area). *Left*: parity plot (predicted vs. true) on the GalaxiesML test set with KDE contours. *Middle*: parity plot for the DDPM-generated set (CNN-predicted vs. SEP-measured). *Upper right*: mean |pred − true| per redshift bin for real (blue) and DDPM (red) images. *Lower right*: difference in mean absolute error per redshift bin (green = DDPM lower error; red = DDPM higher error).

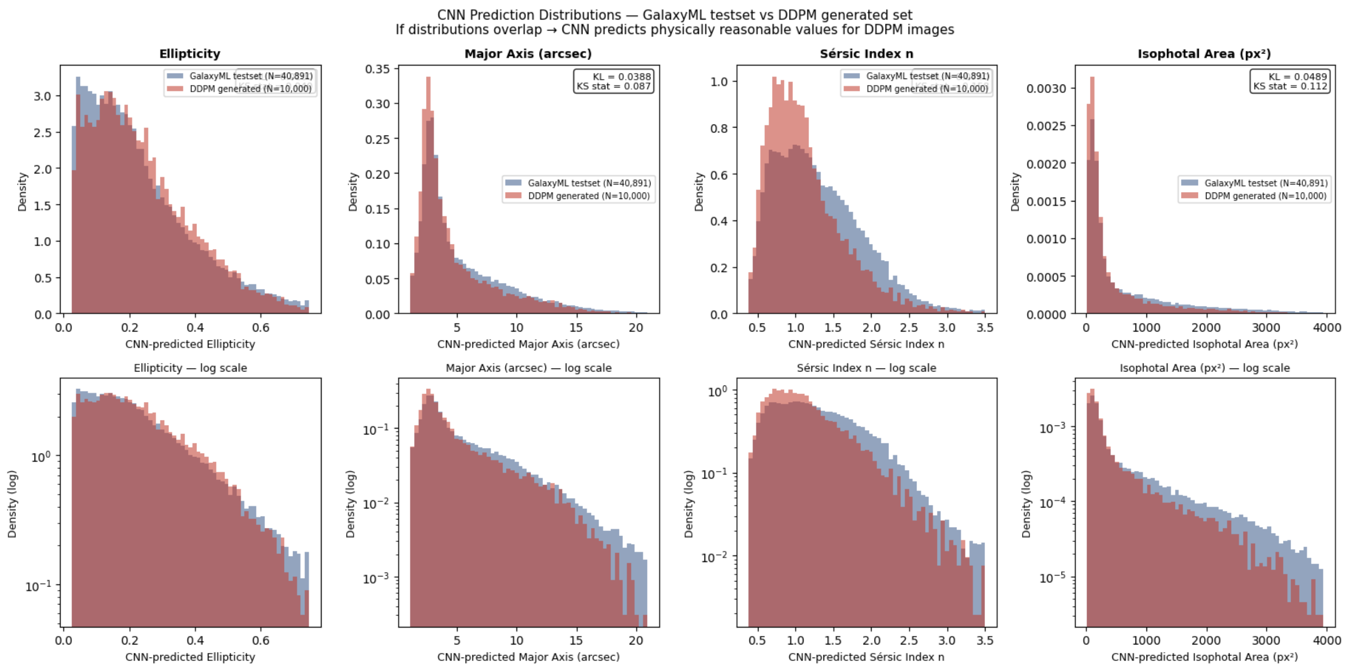

Figure 11. MorphCNN-predicted distributions for the GalaxiesML test set (blue; $N = 40{,}891$) and DDPM-generated images (red; $N = 10{,}000$) across all four morphological parameters. The predicted distributions match closely.

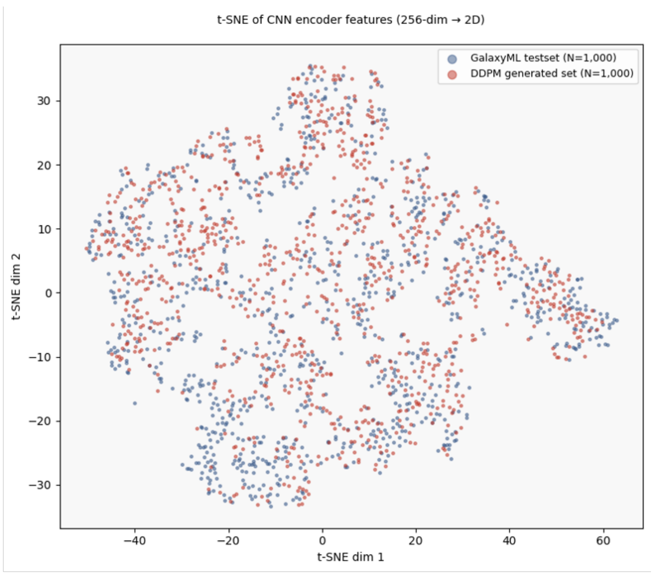

Figure 12. t-SNE projection of 256-dimensional MorphCNN encoder features for 1,000 randomly sampled GalaxiesML test images (blue) and 1,000 DDPM-generated images (red). The two populations mix in feature space with no clear separation, representing the CNN encoder does not perceive a significant structural difference between the two datasets.

other dataset-level differences between the real and generated sets. We discuss further in Section 6.

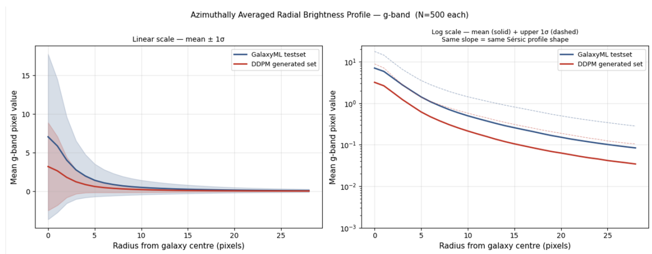

Figure 13. Azimuthally averaged radial brightness profiles in the g band for $N = 500$ randomly sampled GalaxiesML test images (blue) and DDPM-generated images (red). At each radius r, the brightness is averaged over all azimuthal directions (i.e. all pixels at distance r from the galaxy centre), collapsing the 2D image to a 1D curve and reducing noise. Shaded bands show $\pm 1\sigma$. DDPM images are systematically fainter at all radii, while the profile slope is similar, indicating a uniform brightness offset rather than a structural difference in galaxy morphology.

5.7 Redshift Label Perturbation σ and FID

The redshift label perturbation σ (equation 6) controls how much Gaussian noise is added to the conditioning redshift during training. We train DDPM models at $\sigma \in \{0.01, 0.1, 0.5, 1.0\}$ to investigate its effect on generation quality as measured by FID, and adopt $\sigma = 0.01$ as our default based on the results.

Table 4 summarises the training outcomes and FID scores for each configuration. All four models converged successfully.

The results reveal a clear monotonic trend: smaller σ yields lower FID, with scores decreasing from 24.0 at $\sigma = 1.0$ to 15.2 at $\sigma = 0.5$, 14.2 at $\sigma = 0.1$, and 4.96 at $\sigma = 0.01$. This suggests that less perturbation of the conditioning redshift during training allows the model to learn a sharper, more faithful mapping from redshift to galaxy morphology, resulting in higher fidelity generated images.

Table 4. DDPM models trained at different redshift label perturbation strengths σ, with training outcomes and FID scores (lower FID is better). The FID for $\sigma = 0.01$ is our own evaluation on 10,000 generated images against the GalaxiesML test set. FID values marked † are taken from Lizarraga et al. (2024), as our own evaluation could not be completed due to CSD3 cluster downtime; the corresponding training runs completed successfully.

σ	Stopped at epoch	Val loss	FID
0.01	—	—	4.96
0.1	261	0.0005	14.2†
0.5	145	0.0008	15.2†
1.0	263	0.0005	24.0†

† FID from Lizarraga et al. (2024).

In other words, the DDPM benefits from receiving a precise redshift signal rather than a smoothed one. This is consistent with the intuition introduced in Section 4.1.3: as the label perturbation decreases, the conditioning signal becomes sharper, and the model can more easily distinguish fine-grained morphological differences between different redshifts.

6 DISCUSSION

6.1 What the Reproduction Confirmed

Our reproduction of Lizarraga et al. (2024) confirms the core finding that a redshift-conditioned DDPM can generate physically realistic galaxy images. The morphological metric distributions of our generated images match those of the original paper to within 2%(Table 1), and both reproduce the qualitative redshift-dependent trends seen in the real GalaxiesML test set (Fig. 8). The photo-z scatter plot (Fig. 9) further confirms that the DDPM has learned a meaningful mapping from redshift to visual appearance: an independent CNN trained only on real galaxies can recover the conditioned redshift of generated images with $\sigma_{\mathrm{NMAD}} = 0.050$.

6.2 Problems Revealed

6.2.1 *Sérsic index*

We found that the "Sérsic index" reported in Lizarraga et al. (2024) is not a fitted profile parameter but an ellipticity-based proxy (equation 11), a heuristic approximation that does not involve radial brightness profile fitting. We replace this with a proper fitted Sérsic index in our evaluation (Section 4.2.4).

6.2.2 *Measurement Method Mismatch Between SEP and HSC Pipeline*

The GalaxiesML dataset provides pre-computed morphological parameters measured by the HSC photometric pipeline. Although the HSC pipeline also uses SExtractor internally, it applies a different configuration, including different detection thresholds, deblending parameters, and aperture definitions. Since the configuration is unknown, we cannot reproduce and apply it to our DDPM-generated images. To ensure a fair comparison, we therefore re-extracted all four metrics from the GalaxyML test set images using our own sep configuration, identical to what we applied to the generated images.

However, this re-extraction introduces a systematic offset between the SEP-measured values and the HSC catalogue values. Figure 14 shows the correlation between the two measurement methods on the GalaxyML test set. Pearson correlation coefficients range from $r = 0.250$ for ellipticity to $r = 0.635$ for semi-major axis. The fitted Sérsic index achieves a moderate correlation of $r = 0.405$, despite both methods using profile fitting; the remaining scatter reflects differences in the specific fitting procedure, isophote extraction, and sky-subtraction between our SEP-based pipeline and the HSC approach.

As a consequence, if one were to compare our generated-image metrics (measured with SEP) against the HSC catalogue values for real images, apparent distribution mismatches would arise from this methodological difference rather than from any failure of the DDPM. Figure 15 illustrates this: using HSC catalogue values for GalaxyML test set alongside SEP values for generated galaxies produces visibly different distributions, even though the underlying images are the same. This is a difficulty in galaxy morphology studies: different measurement pipelines produce different parameters, and we can not know which one is correct or the best. This motivates our choice to re-extract all metrics with a single consistent tool.

6.2.3 *CNN Redshift Predictor Bias Towards Low Redshift*

As noted in Section 5.5, the CNN predictor over-predicts at low redshift for the generated images. Moreover, the mean $|\Delta z/(1+z)|$ increases monotonically with redshift (Fig. 9, top right), indicating that prediction uncertainty grows with z and the model becomes progressively less reliable at high redshift. A horizontal stripe at $z \approx 2.7$ further suggests that the model collapses to a fixed prediction for some high-redshift inputs.

Both effects are rooted in the GalaxiesML training data distribution (Fig. 16): the dataset is strongly dominated by low-redshift galaxies, with very few examples at $z > 2$. As a result, the CNN predictor is biased towards the dominant low-z class and has seen too few high-z examples to learn reliable features at those redshifts. The DDPM is trained on the same imbalanced distribution, so its generated images at high redshift may not faithfully reflect the true diversity of high-z morphologies, further compounding the predictor bias.

6.2.4 *MorphCNN performance gap reflection*

A plausible explanation for the systematic brightness deficit and SNR difference reflected in 5.6.4 lies in how diffusion models learn the training data distribution. Galaxy images are dominated by dark background pixels: the bright galaxy core occupies only a small fraction of the 64×64 image, while the vast majority of pixels contain sky background at near-zero flux. The DDPM is therefore trained on a pixel distribution that is heavily skewed towards low values. During generation, the model implicitly averages over this distribution, which is a phenomenon known as **regression to the mean**, producing images in which the galaxy core is systematically underexposed relative to real HSC observations. As a result, the generated images are darker than the images in GalaxyML test set. This is a limitation of diffusion models on datasets with highly imbalanced pixel distributions, and is not specific to galaxy imaging.

A further source of the performance gap is a label inconsistency between training and evaluation. MorphCNN was trained on the HSC photometric catalogue values stored in the GalaxiesML dataset, which were derived from the HSC pipeline. When evaluating on DDPM-generated images, we obtained the "ground truth" by running SEP on each generated image. As demonstrated in Section 6.2.2, the HSC pipeline and SEP do not produce equivalent measurements for the same galaxy, so the model is effectively being compared against a different measurement system to the one it was trained on. Retraining MorphCNN on SEP-extracted labels for the full 204,573-image training set would eliminate this inconsistency, but this task could not be completed due to unexpected downtime of the CSD3 HPC cluster.

6.3 Limitations and Future Work

6.3.1 *Compute Interruptions on CSD3*

Most training, evaluation, and generation tasks were conducted on the CSD3 HPC cluster. However, several jobs were interrupted mid-run by unexpected cluster downtime, leaving key tasks incomplete. Specifically, the FID evaluation for the $\sigma \in \{0.1, 0.5, 1.0\}$ ablation models (Section 5.7) was terminated before completion. Also, the SEP metric extraction across the full 204,573-image GalaxiesML training set was also terminated, which would have enabled retraining MorphCNN on consistent SEP-based labels Section 5.6. Other planned experiments also could not be completed within the available compute window.

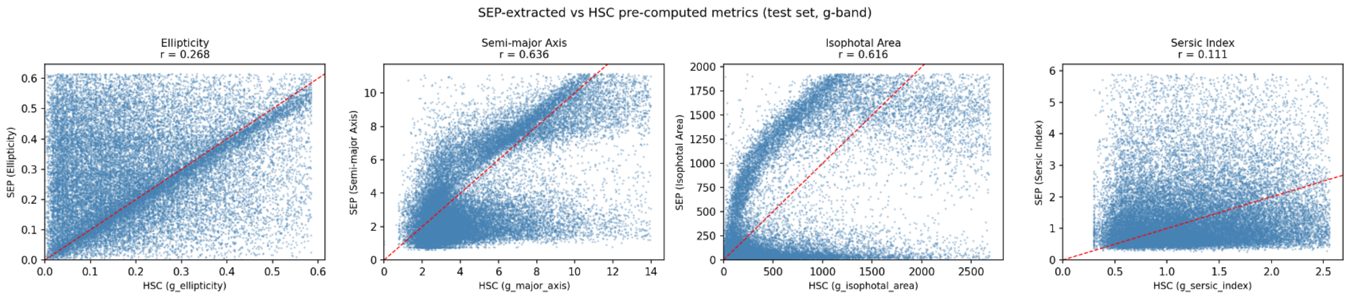

Figure 14. Per-galaxy comparison of SEP-extracted metrics (vertical axis) versus HSC pre-computed catalogue values (horizontal axis) for the GalaxiesML test set, measured in the g band. The Sérsic index shown is our fitted value from isophote profile fitting (Section 4.2.4), not the ellipticity proxy. Pearson correlation coefficients are shown for each metric. The moderate correlation for the fitted Sérsic index ($r = 0.405$) reflects methodological differences between our isophote-based 1D profile fit and the HSC pipeline's approach, despite both using profile fitting.

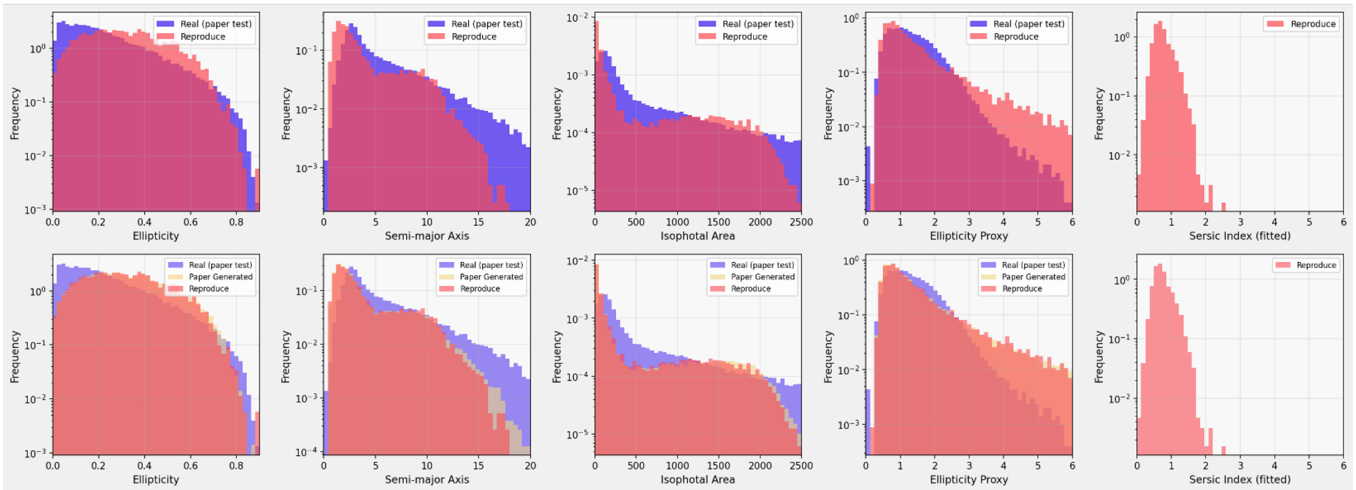

Figure 15. Morphological metric distributions when HSC catalogue values are used for real galaxies (blue) rather than SEP re-extracted values. *Above*: comparison between HSC catalogue metrics (real) and SEP-extracted metrics (reproduced DDPM). *Below*: three-way comparison additionally including the original paper's generated images. The distribution mismatches become apparent due to the measurement method difference.

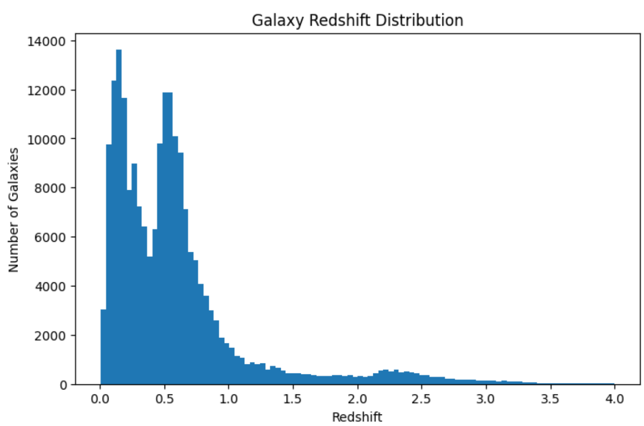

Figure 16. Redshift distribution of the GalaxiesML training set. The dataset is strongly skewed towards low redshift, with the majority of galaxies at $z < 1$. High-redshift galaxies ($z > 2$) are sparsely represented, which biases both the CNN predictor and the DDPM towards low-redshift features.

6.3.2 High-Redshift Data Sparsity and Single-Band Metric Extraction

A fundamental limitation of this work is the severe class imbalance in the GalaxiesML dataset (Fig. 16). Because high-redshift galaxies ($z > 2$) are sparsely represented in both the training and test sets, neither the DDPM nor the CNN predictor can be adequately evaluated at high redshift. Future work could address this by applying redshift-stratified sampling or augmentation strategies during training to balance the class distribution.

6.3.3 Single-Band Metric Extraction and Multi-Band Conditioning

A further limitation concerns how the five HSC bands are used. Although the DDPM and MorphCNN both take all five bands as input, the SEP-based evaluation metrics (ellipticity, semi-major axis, Sérsic index, and isophotal area) are extracted from the g band only, discarding the morphological information encoded in the remaining four bands. Similarly, the visual comparisons render the five-band images as three-channel RGB composites (mapping $i \to R$, $r \to G$, $g \to B$), omitting the z and y bands entirely. These choices are common in astrophysics and save computing resources, but discard potentially useful signal, and future work should extract metrics from all five bands independently.

More fundamentally, the current DDPM is conditioned only on redshift. This work shows that a CNN predictor can successfully back-predict redshift from generated images precisely compared than the MorphCNN. One of the reasons is that redshift is the conditioning variable, so the model is explicitly trained to embed redshift information into the generated images. By the same logic, MorphCNN struggles to recover morphological parameters from generated images because those parameters were never part of the conditioning signal. A natural extension would be to train a DDPM conditioned jointly on redshift *and* morphological parameters (e.g. ellipticity, Sérsic index), analogously to how text-to-image diffusion models accept multiple conditioning inputs. Such a model would allow controlled generation of galaxies with specified morphologies, and would provide a stronger test of whether the diffusion model has truly learned the relationship between physical parameters and visual appearance.

7 CONCLUSIONS

We reproduced the redshift-conditioned DDPM of Lizarraga et al. (2024) and confirmed its core findings. The DDPM successfully learns to generate physically realistic galaxy images conditioned on redshift, reproducing observed morphological distributions and colour evolution trends. The CNN redshift predictor further confirms this: an independent model trained only on real galaxies can recover the conditioned redshift of generated images with reasonable accuracy, demonstrating that the generated images carry meaningful physical information.

We introduced several extensions. First, we fit a proper Sérsic index to replace the original ellipticity proxy. Second, we propose MorphCNN, a residual CNN for predicting four morphological parameters (ellipticity, semi-major axis, Sérsic index, and isophotal area) directly from galaxy images. Third, we also introduced new training strategies, including early stopping and a learning rate scheduler, to improve DDPM training efficiency. Finally, we provide several analytical improvements, including per-bin residual analysis of CNN redshift predictions, colour evolution analysis of generated samples, and a systematic comparison of different morphological measurement approaches including the effect of redshift label perturbation σ on generation quality.

Several problems were also identified. SEP and the HSC pipeline produce systematically different measurements for the same galaxies, and this mismatch must be accounted for in any fair comparison between generated and real images. MorphCNN performs substantially worse on generated images than on real ones. We attribute this to a systematic brightness deficit in generated images (regression to the mean) and a label inconsistency between the measurement systems used for training and evaluation.

Future work should address high-redshift data sparsity through redshift-stratified sampling, extend metric extraction to all five HSC bands, and explore conditioning the DDPM jointly on redshift and morphological parameters to enable controlled galaxy generation.

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DATA AVAILABILITY

The GalaxiesML dataset used in this work is publicly available on Zenodo at https://zenodo.org/records/11117528.

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